

The noise magnitude depends in many cases on the signal magnitude itself

$$f = g \nu. \quad (2.16)$$

This model describes **multiplicative noise**. An example of multiplicative noise is television raster degradation, which depends on TV lines; in the area of a line this noise is maximal, and between two lines it is minimal. Another example of multiplicative noise is the degradation of film material caused by the finite size of silver grains used in photosensitive emulsion.

Quantization noise occurs when insufficient quantization levels are used, for example, 50 levels for a monochromatic image. In this case false contours appear. Quantization noise can be eliminated simply, see Section 2.2.2.

Impulse noise means that an image is corrupted with individual noisy pixels whose brightness differs significantly from that of the neighborhood. The term **salt-and-pepper noise** is used to describe saturated impulsive noise—an image corrupted with white and/or black pixels is an example. Salt-and-pepper noise can corrupt binary images.

The problem of suppressing noise in images is addressed in Chapter 5. If nothing is known a priori about noise properties, local pre-processing methods are appropriate (Section 5.3). If the noise parameters are known in advance, image restoration techniques can be used (Section 5.4).

2.4 Color images

Human color perception adds a subjective layer on top of underlying objective physical properties—the wavelength of electromagnetic radiation. Consequently, color may be considered a psychophysical phenomenon.

Color has long been used in painting, photography and films to display the surrounding world to humans in a similar way to that in which it is perceived in reality. There is considerable literature on the variants in the naming of colors across languages, which is a very subtle affair [Kay, 2005]. The human visual system is not very precise in perceiving color in absolute terms; if we wish to express our notion of color precisely we would describe it relative to some widely used color which is used as a standard: recall, e.g., the red of a British public telephone box. There are whole industries which present images to humans—the press, films, displays, and hence a desire for color constancy. In computer vision, we have the advantage of using a camera as a measuring device, which yields measurements in absolute quantities.

Newton reported in the 17th century that white light from the sun is a spectral mixture, and used the optical prism to perform decomposition. This was a radical idea to propose at time; over 100 years later influential scientists and philosophers such as Goethe refused to believe it.

2.4.1 Physics of color

The electromagnetic spectrum is illustrated in Figure 2.23.

Only a narrow section of the electromagnetic spectrum is visible to a human, with wavelength λ from approximately 380 nm to 740 nm. Visible colors with the wavelengths shown in Figure 2.24 are called **spectral colors** and are those which humans see when white light is decomposed using a Newtonian prism, or which are observed in a rainbow

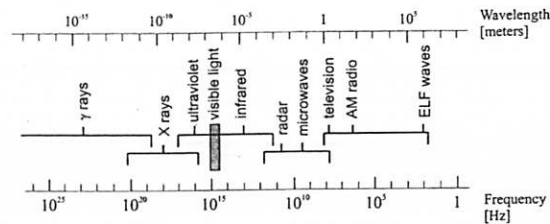


Figure 2.23: Division of the whole electromagnetic spectrum (ELF means Extremely Low Frequencies).

on the sky. Colors can be represented as combinations of the **primary colors**, e.g., red, green, and blue, which for the purposes of standardization have been defined as 700 nm, 546.1 nm, and 435.8 nm, respectively [Pratt, 1978], although this standardization does not imply that all colors can be synthesized as combinations of these three.

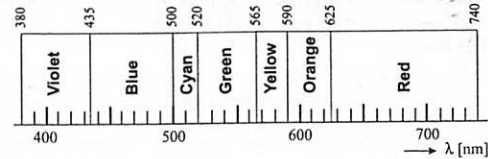


Figure 2.24: Wavelength λ of the spectrum visible to humans.

The intensity of irradiation for different wavelengths λ usually changes. This variation is expressed by a **power spectrum** (called also power spectrum distribution) $S(\lambda)$.

Why do we see the world in color? There are two predominant physical mechanisms describing what happens when a surface is irradiated. First, the **surface reflection** rebounds incoming energy in a similar way to a mirror. The spectrum of the reflected light remains the same as that of the illuminant and it is independent of the surface—recall that shiny metals 'do not have a color'. Second, the energy diffuses into the material and reflects randomly from the internal pigment in the matter. This mechanism is called **body reflection** and is predominant in dielectrics as plastic or paints. Figure 2.25 illustrates both surface reflection (mirroring along surface normal $\mathbf{n}$) and body reflection. Colors are caused by the properties of pigment particles which absorb certain wavelengths from the incoming illuminant wavelength spectrum.

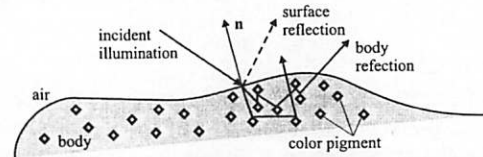


Figure 2.25: Observed color of objects is caused by certain wavelength absorptions by pigment particles in dielectrics.

Most sensors used for color capture, e.g., in cameras, do not have direct access to color; the exception is a **spectrophotometer** which in principle resembles Newton's prism. Incoming irradiation is decomposed into spectral colors and intensity along the spectrum with changing wavelength λ is measured in a narrow wavelength band, for instance, by a mechanically moved point sensor. Actual spectrophotometers use diffraction gratings instead of a glass prism.

Sometimes, intensities measured in several narrow bands of wavelengths are collected in a vector describing each pixel. Each spectral band is digitized independently and is represented by an individual digital image function as if it were a monochromatic image. In this way, **multispectral images** are created. Multispectral images are commonly used in remote sensing from satellites, airborne sensors and in industry. Wavelength usually span from ultraviolet through the visible section to infrared. For instance, the LANDSAT 4 satellite transmits digitized images in five spectral bands from near-ultraviolet to infrared. Seven or a dozen wavelength bands are common.

2.4.2 Color perceived by humans

Evolution has developed a mechanism of indirect color sensing in humans and some animals. Three types of sensors receptive to the wavelength of incoming irradiation have been established in humans, thus the term **trichromacy**. Color sensitive receptors on the human retina are the **cones**. The other light sensitive receptors on the retina are the **rods** which are dedicated to sensing monochromatically in low ambient light conditions. Cones are categorized into three types based on the sensed wavelength range: S (short) with maximum sensitivity at ≈ 430 nm, M (medium) at ≈ 560 nm, and L (long) at ≈ 610 nm. Cones S, M, L are occasionally called cones B, G and R, respectively, but that is slightly misleading. We do not see red solely because an L cone is activated. Light with equally distributed wavelength spectrum looks white to a human, and an unbalanced spectrum appears as some shade of color.

The reaction of a photoreceptor or output from a sensor in a camera can be modeled mathematically. Let i be the specific type of sensor, $i = 1, 2, 3$, (the retinal cone type S, M, L in the human case). Let $R_i(\lambda)$ be the spectral sensitivity of the sensor, $I(\lambda)$ be the spectral density of the illumination, and $S(\lambda)$ describe how the surface patch reflects each wavelength of the illuminating light. The spectral response q_i of the i -th sensor, can be modeled by integration over a certain range of wavelengths

$$q_i = \int_{\lambda_1}^{\lambda_2} I(\lambda) R_i(\lambda) S(\lambda) d\lambda. \quad (2.17)$$

Consider the cone types S, M, L. How does the vector (q_S, q_M, q_L) represent the color or the surface patch? It does not according to equation (2.17) since the output from the photosensors depends on the three factors $I(\lambda)$, $S(\lambda)$ and $R(\lambda)$. Only the factor $S(\lambda)$ is related to the surface patch. Only in the ideal case, when the illumination is perfectly white, i.e., $I(\lambda) = 1$, can we consider (q_S, q_M, q_L) as an estimate of the color of the surface.

Figure 2.26 illustrates qualitatively the relative sensitivities of S, M, L cones. Measurements were taken with the white light source at the cornea so that absorption of wavelength in cornea, lens and inner pigments of the eye is taken into account [Wandell, 1995].

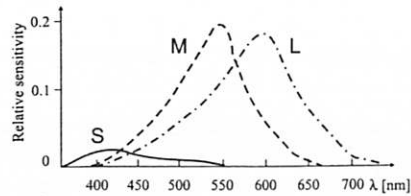


Figure 2.26: Relative sensitivity of S, M, L cones of the human eye to wavelength.

A phenomenon called **color metamer** is relevant. A metamer, in general, means two things that are physically different but perceived as the same. Red and green adding to produce yellow is a color metamer, because yellow could have also been produced by a spectral color. The human visual system is fooled into perceiving that red and green is the same as yellow.

Consider a color matching experiment in which someone is shown a pattern consisting of two adjacent color patches. The first patch displays a test light—a spectral color of certain wavelength. The second patch is created as an additive combination of three selected primary lights, e.g., colors red=645.2 nm, green=525.3 nm and blue=444.4 nm. The observer is asked to control the red, green and blue intensities until both patches look identical. This color matching experiment is possible because of the color metamer. The result of measurements (redrawn from [Wandell, 1995]) is in Figure 2.27. Negative lobes can be seen on the curves for red and green in this figure. This would seem to be impossible. For wavelengths exhibiting negative values the three additive lights do not perceptually match the spectral color because it is darker. If the perceptual match has to be obtained then the observer has to add the intensity to the patch corresponding to the spectral color. This increase of this intensity is depicted as a decrease in the color matching function. Hence the negative values.

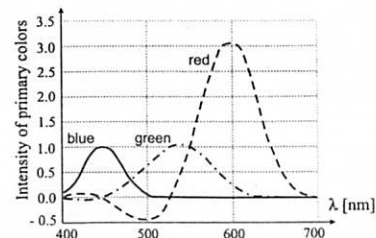


Figure 2.27: Color matching functions obtained in the color matching experiment. Intensities of the selected primary colors which perceptually match spectral color of given wavelength λ .

Human vision is prone to various illusions. Perceived color is influenced, besides the spectrum of the illuminant, by the colors and scene interpretation surrounding the observed color. In addition, eye adaptation to changing light conditions is not very fast and perception is influenced by adaptation. Nevertheless, we assume for simplicity that the spectrum of light coming to a point on the retina fully determines the color.

Since color can be defined by almost any set of primaries, the world community agreed on primaries and color matching functions which are widely used. The **color model** was introduced as a mathematical abstraction allowing us to express colors as tuples of numbers, typically as three or four values of color components. Being motivated by the press and the development of color film, in 1931, CIE (International Commission on

Illumination, still acting in Lausanne, Switzerland) issued a technical standard called **XYZ color space**.

The standard is given by the three imaginary lights $X=700.0$ nm, $Y=546.1$ nm, $Z=435.8$ nm and by the color matching functions $X(\lambda)$, $Y(\lambda)$ and $Z(\lambda)$ corresponding to the perceptual ability of an average human viewing a screen through an aperture providing a 2° field of view. The standard is artificial because there is no set of physically realizable primary lights that would yield the color matching functions in the color matching experiment. Nevertheless, if we wanted to characterize the imaginary lights then, very roughly speaking, $X \approx$ red, $Y \approx$ green and $Z \approx$ blue. The CIE standard is an example of an absolute standard, i.e., defining unambiguous representation of color which does not depend on other external factors. There are more recent and more precise absolute standards: CIELAB 1976 (ISO 13665) and HunterLab (<http://www.hunterlab.com>). Later, we will also discuss relative color standards such as RGB color space. There are several RGB color spaces used—two computer devices may display the same RGB image differently.

The XYZ color standard fulfills three requirements:

- Unlike the color matching experiment yielding negative lobes of color matching functions, the color matching functions of XYZ color space are required to be non-negative;
- The value of $Y(\lambda)$ should coincide with the brightness (luminance);
- Normalization is performed to assure that the power corresponding to the three color matching functions is equal (i.e., the area under all three curves is equal).

The resulting color matching functions are shown in Figure 2.28. The actual color is a mixture (more precisely a convex combination) of

$$c_X X + c_Y Y + c_Z Z, \quad (2.18)$$

where $0 \leq c_X, c_Y, c_Z \leq 1$ are weights (intensities) in the mixture. The subspace of colors perceivable by humans is called the color gamut and is demonstrated in Figure 2.29.

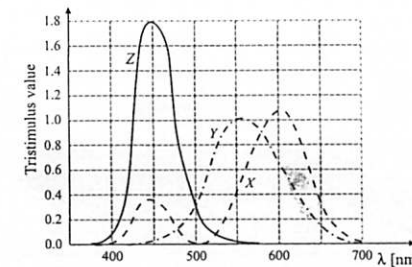


Figure 2.28: Color matching functions for the CIE standard from 1931. $X(\lambda)$, $Y(\lambda)$, $Z(\lambda)$ are color matching functions. Redrawn from [Wandell, 1995].

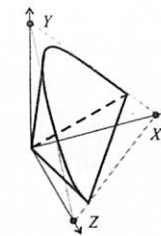


Figure 2.29: Color gamut - a subspace of the X, Y, Z color space showing all colors perceivable by humans.

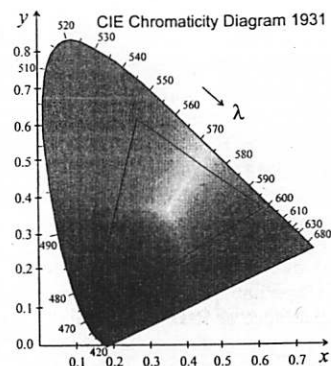


Figure 2.30: CIE chromaticity diagram is a projection of XYZ color space into a plane. The triangle depicts a subset of colors spanned by red, green, and blue. These are TV colors, i.e., all possible color which can be seen on a CRT display. A color version of this figure may be seen in the color inset—Plate 1.

3D figures are difficult to handle in publications, and so a planar view of a 3D color space is used. The projection plane is given by the plane passing through extremal points on all three axes, i.e., points X, Y, Z . The new 2D coordinates x, y are obtained as

$$x = \frac{X}{X+Y+Z}, \quad y = \frac{Y}{X+Y+Z}, \quad z = 1 - x - y.$$

The result of this plane projection is the CIE chromaticity diagram, see Figure 2.30. The horseshoe like subspace contains colors which people are able to see. All monochromatic spectra visible to humans map into the curved part of the horseshoe—their wavelengths are shown in Figure 2.30.

Display and printing devices use three selected real primary colors (as opposed to three syntactic primary colors of XYZ color space). All possible mixtures of these primary colors fail to cover the whole interior of the horseshoe in CIE chromaticity diagram. This situation is demonstrated qualitatively for three particular devices in Figure 2.31.

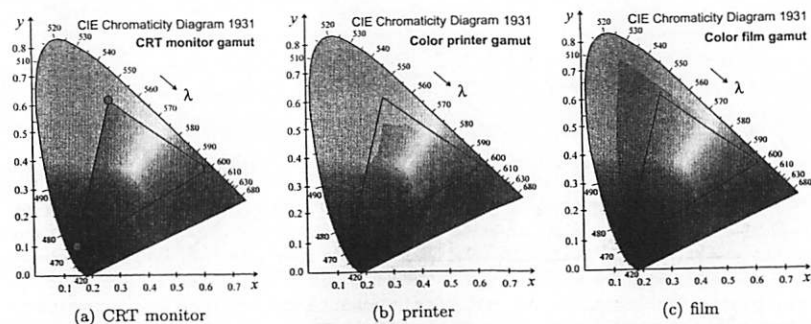


Figure 2.31: Gamuts which can be displayed using three typical display devices. A color version of this figure may be seen in the color inset—Plate 2.

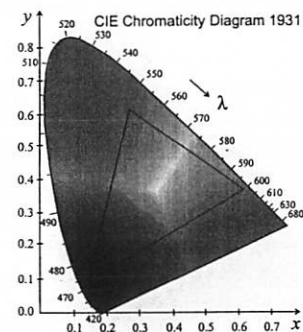


Plate 1: Page 36, Figure 2.30.

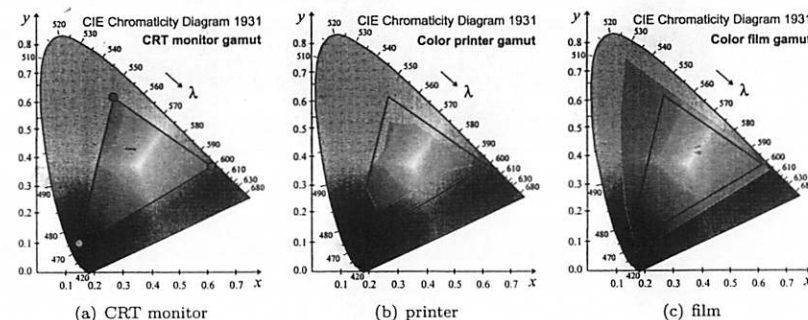


Plate 2: Page 36, Figure 2.31.

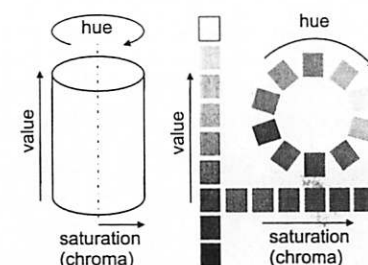


Plate 3: Page 38, Figure 2.33.

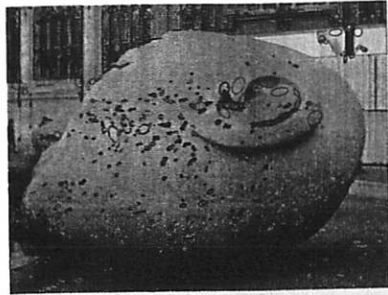


Plate 16: Page 647, Figure 12.30.

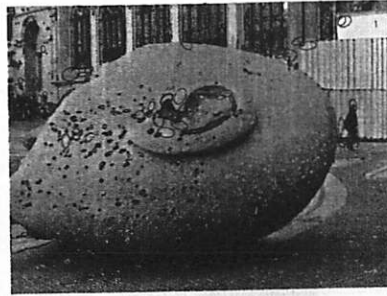


Plate 17: Page 647, Figure 12.31.

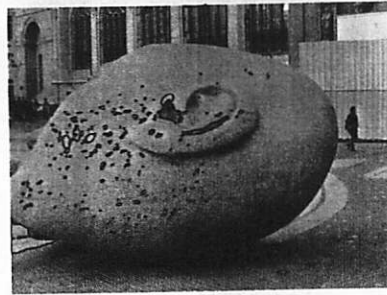
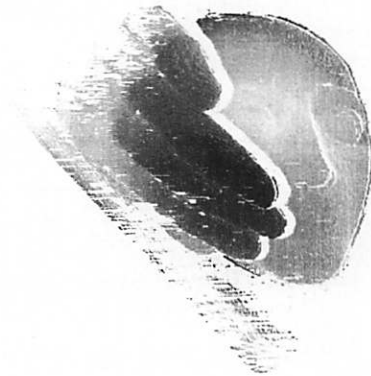


Plate 18: Page 649, Figure 12.34.



2.4.3 Color spaces

Several different primary colors and corresponding color spaces are used in practice, and these spaces can be transformed into each other. If the absolute color space is used then the transformation is the one-to-one mapping and does not lose information (except for rounding errors). Because color spaces have their own gamuts, information is lost if the transformed value appears out of the gamut. See [Burger and Burge, 2006] for a full explanation and for algorithms; here, we list several frequently used color spaces.

The RGB color space has its origin in color television where Cathode Ray Tubes (CRT) were used. RGB color space is an example of a relative color standard (as opposed to the absolute one, e.g., CIE 1931). The primary colors (R-red, G-green and B-blue) mimicked phosphor in CRT luminophore. The RGB model uses additive color mixing to inform what kind of light needs to be emitted to produce a given color. The value of a particular color is expressed as a vector of three elements—intensities of three primary colors, recall equation (2.18). A transformation to a different color space is expressed by a transformation by a 3×3 matrix. Assume that values for each primary are quantized to $m = 2^n$ values; let the highest intensity value be $k = m - 1$; then $(0, 0, 0)$ is black, (k, k, k) is (television) white, $(k, 0, 0)$ is 'pure' red, and so on. The value $k = 255 = 2^8 - 1$ is common, i.e., 8 bits per color channel. There are $256^3 = 2^{24} = 16,777,216$ possible colors in such a discretized space.

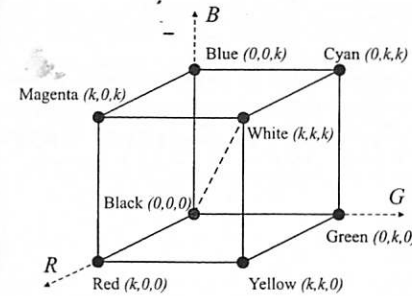


Figure 2.32: RGB color space with primary colors red, green, blue and secondary colors yellow, cyan, magenta. Gray-scale images with all intensities lie along the dashed line connecting black and white colors in RGB color space.

The RGB model may be thought of as a 3D co-ordinatization of color space (see Figure 2.32); note the secondary colors which are combinations of two pure primaries. There are specific instances of the RGB color model as sRGB, Adobe RGB and Adobe Wide Gamut RGB. They differ slightly in transformation matrices and the gamut. One of transformations between RGB and XYZ color spaces is

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 3.24 & -1.54 & -0.50 \\ -0.98 & 1.88 & 0.04 \\ 0.06 & -0.20 & 1.06 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \quad (2.19)$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.41 & 0.36 & 0.18 \\ 0.21 & 0.72 & 0.07 \\ 0.02 & 0.12 & 0.95 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}.$$

The US and Japanese color television formerly used YIQ color space. The Y component describes intensity and I, Q represent color. YIQ is another example of

additive color mixing. This system stores a luminance value with two chrominance values, corresponding approximately to the amounts of blue and red in the color. This color space corresponds closely to the YUV color model in the PAL television norm (Australia, Europe, except France, which uses SECAM). YIQ color space is rotated 33° with respect to the YUV color space. The YIQ color model is useful since the Y component provides all that is necessary for a monochrome display; further, it exploits advantageous properties of the human visual system, in particular our sensitivity to luminance, the perceived energy of a light source.

The CMY—for Cyan, Magenta, Yellow—color model uses subtractive color mixing which is used in printing processes. It describes what kind of inks need to be applied so the light reflected from the white substrate (paper, painter's canvas) and passing through the inks produces a given color. CMYK stores ink values for black in addition. Black color can be generated from C, M, Y components but as it is abundant in printed documents, it is of advantage to have a special black ink. Many CMYK colors spaces are used for different sets of inks, substrates, and press characteristics (which change the color transfer function for each ink and thus change the appearance).

HSV—Hue, Saturation, and Value (also known as HSB, hue, saturation, brightness) is often used by painters because it is closer to their thinking and technique. Artists commonly use three to four dozen colors (characterized by the hue; technically, the dominant wavelength). If another color is to be obtained then it is mixed from the given ones, for example, 'purple' or 'orange'. The painter also wants colors of different saturation, e.g., to change 'fire brigade red' to pink. She will mix the 'fire brigade red' with white (and/or black) to obtain the desired lower saturation. The HSV color model is illustrated in Figure 2.33.

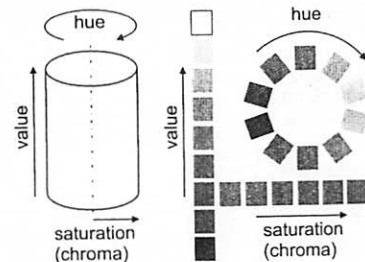


Figure 2.33: HSV color model illustrated as a cylinder and unfolded cylinder. A color version of this figure may be seen in the color inset—Plate 3.

HSV decouples intensity information from color, while hue and saturation correspond to human perception, thus making this representation very useful for developing image processing algorithms. This will become clearer as we proceed to describe image enhancement algorithms (for example, equalization Algorithm 5.1), which if applied to each component of an RGB model would corrupt the human sense of color, but which would work more or less as expected if applied to the intensity component of HSV (leaving the color information unaffected). HSL (hue, saturation, lightness/luminance), also known as HLS or HSI (hue, saturation, intensity) is similar to HSV. 'Lightness' replaces 'brightness'. The difference is that the brightness of a pure color is equal to the brightness of white, while the lightness of a pure color is equal to the lightness of a medium gray.

Models	Color spaces	Applications
Colorimetric	XYZ	Colorimetric calculations
Device oriented, nonuniform spaces	RGB, UIQ	Storage, processing, coding, color TV
Device oriented, Uniform spaces	LAB, LUV	Color difference, analysis
User oriented	HSL, HSI	Color perception, computer graphics

2.4.4 Palette images

Palette images (called also indexed images) provide a simple way to reduce the amount of data needed to represent an image. The pixel values constitute a link to a lookup table (also called a color table, color map, index register, palette). The lookup table contains as many entries as the range of possible values in the pixel, which is typically 8 bits $\equiv$ 256 values. Each entry of the table maps the pixel value to the color, so there are three values, one for each of three color components. In the typical case of the RGB color model, values for red, green and blue are provided. It is easy to see that this approach would reduce data consumption to one-third if each of the RGB channels had been using 8 bits (plus size of the look up table). Many widely used image formats for raster images such as TIFF, PNG and GIF can store palette images.

If the number of colors in the input image is less than or equal to the number of entries in the lookup table then all colors can be selected and no loss of information occurs. Such images may be cartoon movies, or program outputs. In the more common case, the number of colors in the image exceeds the number of entries in the lookup table and a subset of colors has to be chosen, and a loss of information occurs.

This color selection may be done many ways. The simplest is to quantize color space regularly into cubes of the same size. In the 8 bit example, there would be $8 \times 8 \times 8 = 256$ such cubes. If there is, e.g., a green frog in green grass in the picture then there will not be enough shades of green available in the lookup table to display the image well. In such a case, it is better to check which colors appear in the image by creating histograms for all three color components and quantize them to provide more shades for colors which occur in the image frequently. If an image is converted to a palette representation then the nearest color (in some metric sense) in the lookup table is used to represent the color. This is an instance of **vector quantization** (see Section 14.4) which is widely used in analyzing large multi-dimensional datasets. It is also possible to view the occupation by the pixels of RGB space as a **cluster analysis** problem (see Section 9.2.5), susceptible to algorithms such as k-means (Algorithm 9.4).

The term **pseudocolor** is usually used when an original image is gray-level and is displayed in color; this is often done to exploit the color discriminatory power of human vision. The same palette machinery as described above is used for this purpose; a palette is loaded into the lookup table which visualizes the particular gray-scale image the best. It could either enhance local changes, or might provide various views of the image. Which palette to choose depends on the semantics of the image and cannot be derived from image statistics alone. This selection is an interactive process.

Almost all computer graphics cards work with palette images directly in hardware. The content of the lookup table will be filled by the programmer.

2.4.5 Color constancy

Consider the situation in which the same surface is seen under different illumination, e.g., for a Rubik's cube in Figure 2.34. The same surface colors are shown fully illuminated and in shadow. The human vision system is able to abstract to a certain degree from the illumination changes and perceive several instances of a particular color as the same. This phenomenon is called color constancy. Of course, it would be desirable to equip artificial perception systems based on photosensors with this ability too, but this is very challenging.

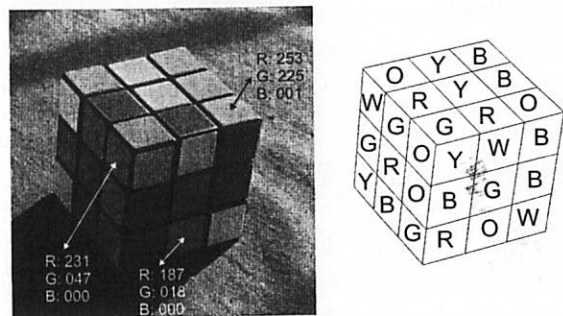


Figure 2.34: Color constancy: The Rubik cube is captured in sunlight, and two of three visible sides of the cube are in shadow. The white balance was set in the shadow area. There are six colors on the cube: R-red, G-green, B-blue, O-orange, W-white, and Y-yellow. The assignment of the six available colors to 3×9 visible color patches is shown on the right. Notice how different the same color patch can be: see RGB values for the three instances of orange. A color version of this figure may be seen in the color inset—Plate 4.

Recall equation (2.17) which models the spectral response q_i of the i -th sensor by integration over a range of wavelengths as a multiplication of three factors: spectral sensitivity $R_i(\lambda)$ of the sensor $i = 1, 2, 3$, spectral density of the illumination $I(\lambda)$, and surface reflectance $S(\lambda)$. A color vision system has to calculate the vector q_i for each pixel as if $I(\lambda) = 1$. Unfortunately, the spectrum of the illuminant $I(\lambda)$ is usually unknown.

Assume for a while the ideal case in which the spectrum $I(\lambda)$ of the illuminant is known. Color constancy could be obtained by dividing the output of each sensor with its sensitivity to the illumination. Let q'_i be the spectral response after compensation for the illuminant (called von Kries coefficients), $q'_i = \rho_i q_i$, where

$$\rho_i = 1 / \int_{\lambda_1}^{\lambda_2} I(\lambda) R_i(\lambda) d\lambda. \quad (2.20)$$

Partial color constancy can be obtained by multiplying color responses of the three photosensors with von Kries coefficients ρ_i .

In practice, there are several obstacles that make this procedure intractable. First, the illuminant spectrum $I(\lambda)$ is not known; it can only be guessed indirectly from reflections in surfaces. Second, only the approximate spectrum is expressed by the spectral response

q_i of the i -th sensor. Clearly the color constancy problem is ill-posed and cannot be solved without making additional assumptions about the scene.

Several such assumptions have been suggested in the literature. It can be assumed that the average color of the image is gray. In such a case, it is possible to scale the sensitivity of each sensor type until the assumption becomes true. This will result in an insensitivity to the color of the illumination. This type of color compensation is often used in automatic white balancing in video cameras.

Another common assumption is that the brightest point in the image has the color of the illumination. This is true when the scene contains specular reflections which have the property that the illuminant is reflected without being transformed by the surface patch.

The problem of color constancy is further complicated by the perceptual abilities of the human visual system. Humans have quite poor quantitative color memory, and also perform color adaptation. The same color is sensed differently in different local contexts.

2.5 Cameras: an overview

2.5.1 Photosensitive sensors

Photosensitive sensors most commonly found in cameras can be divided into two groups:

Sensors based on photo-emission principles explore the photoelectric effect. An external photon carried in incoming radiation brings enough energy to provoke the emission of a free electron. This phenomenon is exhibited most strongly in metals. In image analysis related applications, it has been used in photomultipliers and vacuum tube TV cameras.

Sensors based on photovoltaic principles became widely used with the development of semiconductors. The energy of a photon causes an electron to leave its valence band and changes to a conduction band. The quantity of incoming photons affects macroscopic conductivity. The excited electron is a source of electric voltage which manifests as electric current; the current is directly proportional to the amount of incoming energy (photons). This phenomenon is explored in several technological elements as a photodiode, an avalanche photodiode (an amplifier of light which has similar behavior from the user's point of view as the photomultiplier; it also amplifies noise and is used, e.g., in night vision cameras), a photoresistor, and Schottky photodiode.

There are two types of semiconductor photoresistive sensors used widely in cameras: CCDs (charge-coupled devices) and CMOS (complementary metal oxide semiconductor). Both technologies were developed in laboratories in the 1960s and 1970s. CCDs became technologically mature in the 1970s and became the most widely used photosensors in cameras. CMOS technology started being technologically mastered from about the 1990s. At the time of writing (2006) neither of these technologies is categorically superior to the other. The outlook for both technologies is good.

In a CCD sensor, every pixel's charge is transferred through just one output node to be converted to voltage, buffered, and sent off-chip as an analog signal. All of the pixel area can be devoted to light capture. In a CMOS sensor, each pixel has its own charge-to-voltage conversion, and the sensor often includes amplifiers, noise-correction,

and digitization circuits, so that the chip outputs (digital) bits. These other functions increase the design complexity and reduce the area available for light capture. The chip can be built to require less off-chip circuitry for basic operation.

The basic CCD sensor element includes a Schottky photodiode and a field-effect transistor. A photon falling on the junction of the photodiode liberates electrons from the crystal lattice and creates holes, resulting in the electric charge that accumulates in a capacitor. The collected charge is directly proportional to the light intensity and duration of its falling on the diode. The sensor elements are arranged into a matrix-like grid of pixels—a CCD chip. The charges accumulated by the sensor elements are transferred to a horizontal register one row at a time by a vertical shift register. The charges are shifted out in a bucket brigade fashion to form the video signal.

There are three inherent problems with CCD chips.

- The blooming effect is the mutual influence of charges in neighboring pixels. Current CCD sensor technology is able to suppress this problem (anti-blooming) to a great degree.
- It is impossible to address directly individual pixels in the CCD chip because read out through shift registers is needed.
- Individual CCD sensor elements are able to accumulate approximately 30-200 thousands electrons. The usual level of inherent noise of the CCD sensor is on the level of 20 electrons. The signal-to-noise ratio (SNR) in the case of a cooled CCD chip is $SNR = 20 \log(200000/20)$, i.e., the logarithmic noise is approximately 80 dB at best. This causes that the CCD sensor is able to cope with four orders of magnitude of intensity in the best case. This range drops to approximately two orders of magnitude with common uncooled CCD cameras. The range of incoming light intensity variations is usually higher.

Here, current technology does not beat the human eye. Evolution equipped the human eye with the ability to perceive intensity (brightness) in a remarkable range of nine orders of magnitude (if time for adaptation is provided). This range is achieved because the response of the human eye to intensity is proportional logarithmically to the incoming intensity. Nevertheless, among the sensors available, CCD cameras have high sensitivity (are able to see in darkness) and low levels of noise. CCD elements are abundant, also due to widely used digital photo cameras.

The development of semiconductor technology permits the production of matrix-like sensors based on CMOS technology. This technology is used in mass production in the semiconductor industry because processors and memories are manufactured using the same technology. This yields two advantages. The first is that mass production leads to low prices; because of the same CMOS technology, the photosensitive matrix-like element can be integrated to the same chip as the processor and/or operational memory. This opens the door to 'smart cameras' in which the image capture and basic image processing is performed on the same chip.

The advantage of CMOS cameras (as opposed to CCD) is a higher range of sensed intensities (about 4 orders of magnitude), high speed of read-out (about 100 ns) and random access to individual pixels. The disadvantage is a higher level of noise by approximately one degree of magnitude.

2.5.2 A monochromatic camera

The camera consists of the optical system (lens), the photosensitive sensor(s) and electronics which enables the processing of a captured image, and transfer to further processing.

Analog cameras generate a complete TV signal which contains information about light intensity, and horizontal and vertical synchronization pulses allowing row by row display. The frame scan can be with interlaced lines as in ordinary analog TV, which was introduced to reduce image flickering on cathode-ray tube (CRT) screens. A rate of 60 half-frames per second is used in the USA and Japan, and 50 half-frames per second in Europe and elsewhere. The whole image has 525 lines in the USA and Japan, 625 lines in Europe and elsewhere. Analog cameras require a digitizer card (a frame grabber) to be plugged in to the computer.

Analog cameras have problems with jitter which means that two neighboring lines are not aligned properly and 'float' in a statistical manner one against the other. The human eye is insensitive to jitter because it smoothes out the statistical variation. However, jitter causes problems when the camera is used for measurement purposes such as gauging. Non-interlaced analog cameras with an appropriate frame grabber suppress this problem. Non-interlaced cameras do not need to conform to TV norms, and usually provide higher resolution such as 1024×720 pixels. Nowadays, the preferred solution is to use digital cameras in measurement applications.

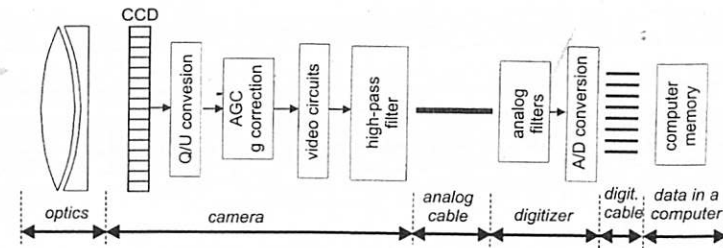


Figure 2.35: Analog CCD camera.

A block diagram of an analog camera with a CCD chip is in Figure 2.35. The block AGC (Automatic Gain Control) automatically changes the gain of the camera according to the amount of light in the scene. The gain is set as a compromise between necessary sensitivity in low illuminated areas and the attempt to avoid saturation of bright areas in the scene.

Cameras usually contain another block named γ correction which performs nonlinear transformation of the intensity scale. The necessity for this in the display chain originates in television technology with cathode-ray tubes (CRT). The dependency between the grid bias voltage U and the resulting irradiance L ($\sim$ brightness) of the luminophore is exponential, $L = U^\beta$ —the typical value of β is 2.2. The shape of the transfer curve is roughly parabolic. Note that the intensity of modern flat Liquid Crystal Displays (LCD) depends linearly on the input voltage.

In order to maintain a linear transfer function of the whole display chain with CRT, there is a need to compensate the non-linear transfer function by an inverse curve. It was

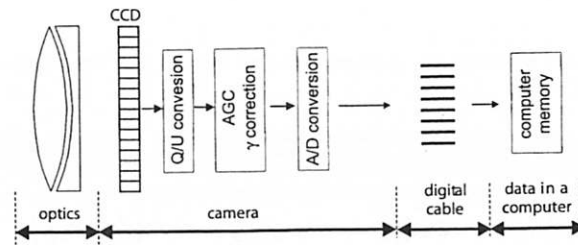


Figure 2.36: Digital CCD camera.

easier and cheaper to include compensation circuitry into a few cameras at the beginning of the television era than to put it into a mass produced TV set. Due to the need for backward compatibility, there is a module in cameras which modifies the dependency of the output voltage of the camera U_k on the input radiation E_k : $U_k = E_k^{1/\beta} = E_k^\gamma$. The typical value is thus $\gamma = 1/2.2 \approx 0.45$. Some cameras allow the γ -value to be set in the range $(0, 1)$. The value $\gamma=1$ corresponds to the correction being switched off.

Sometimes, it is necessary to use the camera as an absolute measuring device of incoming intensities. The image capturing chain has to be radiometrically calibrated before the actual measurement starts. In such a case, there is a need to switch off AGC and γ -correction. Higher quality cameras allow both AGC and γ -correction to be switched on or off. In cheaper cameras, it may be possible to switch off AGC and γ correction by intervening into the camera electronics.

Analog cameras are equipped with video circuitry which adds frame synchronization pulses to the signal. The high-pass filter in the camera compensates for a decrease of high frequencies in the optical part. The TV signal is usually conducted by a coaxial cable to a digitizer (frame grabber) in a computer. At the input of the digitizer, there are sometimes high-pass filters to compensate for the loss of high frequencies in the cable.

A block diagram of a digital camera is in Figure 2.36. The conversion from photon energy to voltage is the same as in analog cameras including potential AGC and/or γ -correction. The analog-to-digital (A/D) converter provides a number proportional to the input intensity. These numbers have to be transferred to the computer for further processing—the connection can be done using either parallel or serial hardware. Serial connections usually exploit widely used technology standardized in the IEEE 1394 (FireWire) or USB (Universal Serial Bus) protocols. In the case of parallel connection, only short cables of length about 1 meter can be used.

Analog cameras	Digital cameras
+ Cheap.	+ Cheap webcams. Dropping price for others.
+ Long cable possible (up to 300 m).	— Shorter cable (≈ 10 m for Firewire). Kilometers after conversion to optical cable. Any length for Internet cameras.
— Multiple sampling of a signal.	+ Single sampling.
— Noisy due to analog transmission.	+ No transmission noise.
— Line jitter.	+ Lines are vertically aligned.

2.5.3 A color camera

Electronic photosensitive sensors are monochromatic. Three strategies are usually employed to capture color images:

- Record three different images in succession by employing color filters in front of monochromatic cameras. This approach is used only in precise laboratory measurements because it is impractical and impossible for any subject involving motion.
- Using a color filter array on a single sensor.
- The incoming light is split into several color channels using a prism-like device.

A color filter array mosaic is often combined with a single photosensitive sensor to create a color camera. Each pixel is covered with an individual filter which can be implemented either on a cover glass on the chip package (hybrid filter) or directly on the silicon (monolithic filter). Each pixel captures only one color. Consequently, the color resolution is about one third of the geometric resolution which would correspond to a monochromatic camera with same number of pixels. The full color values for each pixel can be interpolated from pixel values of the same color in local neighborhood.

The human eye is most sensitive to green, less to red, and least to blue. This property is used by the most common color filter for single chip cameras called the Bayer filter mosaic or pattern (Bryce E. Bayer, US patent dated 1976), see Figure 2.37. It can be seen that the number of green-sensitive pixels is equal to the combined number of pixels sensitive to red and blue.

G	B	G	B	G	B	G	B
R	G	R	G	R	G	R	G
G	B	G	B	G	B	G	B
R	G	R	G	R	G	R	G

Figure 2.37: Bayer filter mosaic for single chip color cameras.

A great advantage of a mosaic filter is its optical simplicity. It provides the single focal plane necessary for the use of standard film lenses. Good mosaic filters provide excellent bandpass transmission. Many professional digital SLR (Single Lens Reflex) and studio cameras use mosaic filters.

Multiple-chip cameras use color filters to split incoming light into separate color channels. Photosensors are simple and preserve the spatial resolution. Aligning and registering the sensors to the color splitter to the prism requires high precision. For pixels of the same size, color splitter systems should allow higher sensitivity in low light conditions, since they should lose less light in the filters. In practice, this advantage is not always available. Color splitting device often include absorption filters because simple refraction may not provide sufficiently precise color separation. The beam splitter approach complicates the optical system and limits lens selection significantly. The additional optical path of the prism increases both lateral and longitudinal aberration for each color's image. The longitudinal aberration causes different focal lengths for each color; the photosensor could be moved independently to each color's focal point, but then the lateral aberration would produce different magnification for each color. These aberrations can be overcome with a lens specifically designed for use with the prism, but such camera-specific lenses would be rare, inflexible, and expensive.

2.6 Summary

• Basic concepts

- A 2D image gray-scale image is represented by a scalar function $f(x, y)$ of two variables which give coordinates in a plane.
- In many cases, a 2D image is formed as the result of a projection of a 3D scene into 2D.
- The domain of the digitized image is a limited discrete grid the coordinates of which are natural numbers. The range of the digitized image is a limited discrete set of gray values (brightnesses). A pixel represents the elemental part of an image.

• Image digitization

- Digitization (sampling) of an image can be seen as a product of a sampling function and a continuous image function.
- Usually the grid consists of regular polygons (squares or hexagons). The second aspect of sampling is setting the distance between the sampling points (the smaller sampling distance the higher the resolution of the image).
- Gray level quantization governs the appearance of shading and false contour. A human is able to recognize about 60 gray levels at most. Images containing only black and white pixels are called the binary.

• Digital image properties

- The neighborhood relation of a pixel has to be defined to be able to represent discrete geometry.
- A function providing distance between two pixels has to be established—there are several definitions used. The most commonly used is ‘city block’, ‘chessboard’, and the Euclidean distance used in everyday life. If the neighborhood relation is set for a grid then a raster is obtained.
- Given a raster, topological properties are induced. These properties are based on the relation ‘being contiguous’ and lead to concepts of region, background, hole, and region border. The convex hull of a region is the minimal convex subset containing it.
- 4-neighborhoods and 8-neighborhoods lead to ‘crossing line’ paradoxes which complicate basic discrete geometry algorithms. However, there exist solutions to these paradoxes for both binary and grey-level images.
- The distance transform (chamfering) of a binary image provides the distance from each pixel to the nearest non-zero pixel. There is a computationally effective two-pass algorithm to compute this, the complexity of which depends linearly on the number of pixels.
- The brightness histogram is a global descriptor of the image giving the estimate of the probability density that a pixel has a given brightness.
- Human visual perception is vulnerable to various illusions. Some of the properties of human perception of images as perceptual grouping are inspirations for computer vision methods.
- Live images as any other measurements or observations are always prone to noise. It is possible to assess the noise extent quantitatively using, e.g., signal-to-noise ratio.

- White, Gaussian, impulse, and salt-and-pepper noise are common models.

• Color images

- Human color perception is a subjective psychophysical layer on top of underlying objective physical properties—the wavelength of electromagnetic radiation.
- Three types of sensors receptive to the wavelength of incoming irradiation have been established in humans. Color sensitive receptors on the human retina are cones. The other light sensitive receptors on the retina are rods which are dedicated to sensing monochromatically in low ambient light conditions. Cones are categorized into three types based on the sensed wavelength range, approximately corresponding to red, green and blue.

• Cameras

- Most cameras use either CCD or CMOS photosensitive elements, both using photovoltaic principles. They capture brightness of a monochromatic image.
- Cameras are equipped with necessary electronics to provide digitized images. Color cameras are similar to monochromatic ones and contain color filters.

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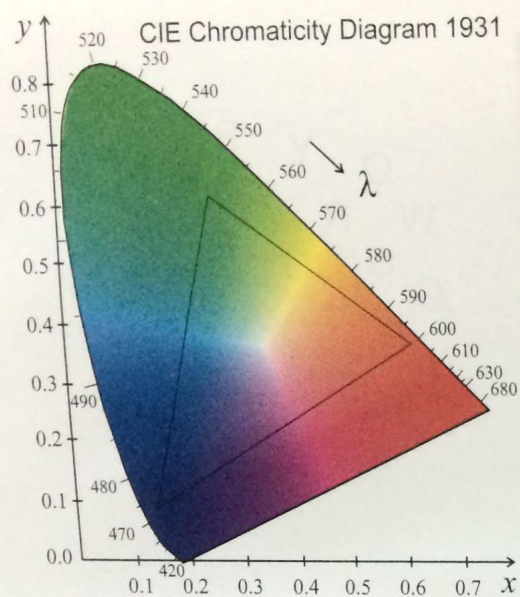


Plate 1: Page 36, Figure 2.30.

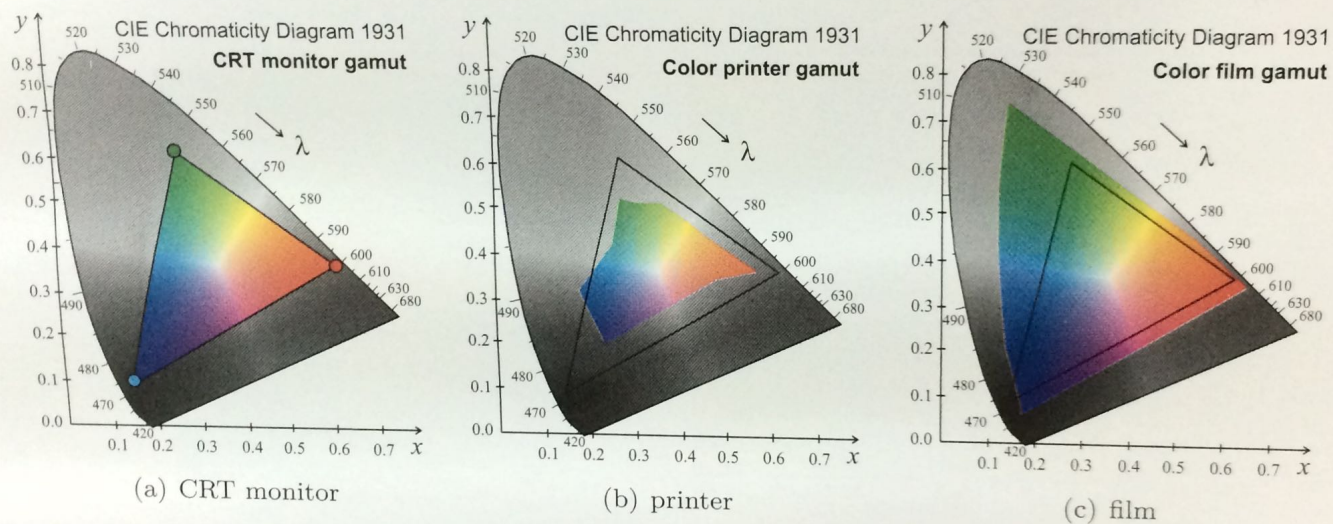


Plate 2: Page 36, Figure 2.31.

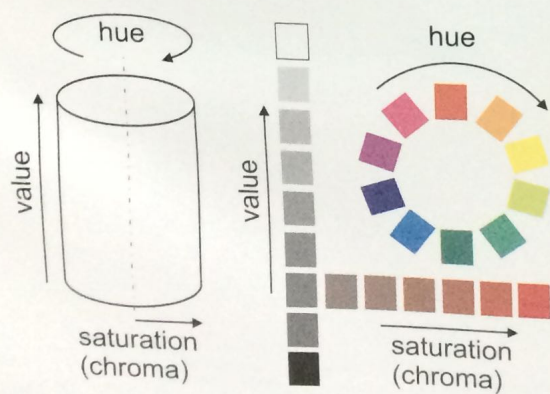


Plate 3: Page 38, Figure 2.33.

B Color inset

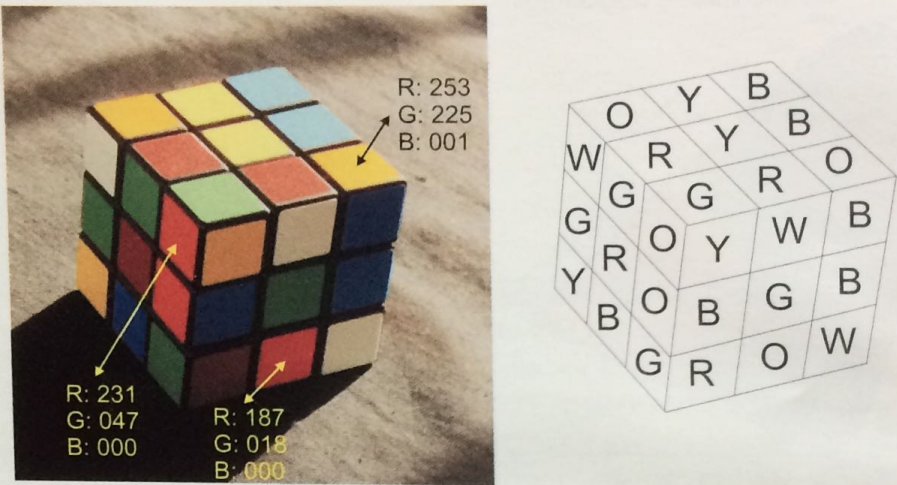


Plate 4: *Page 40, Figure 2.34.*

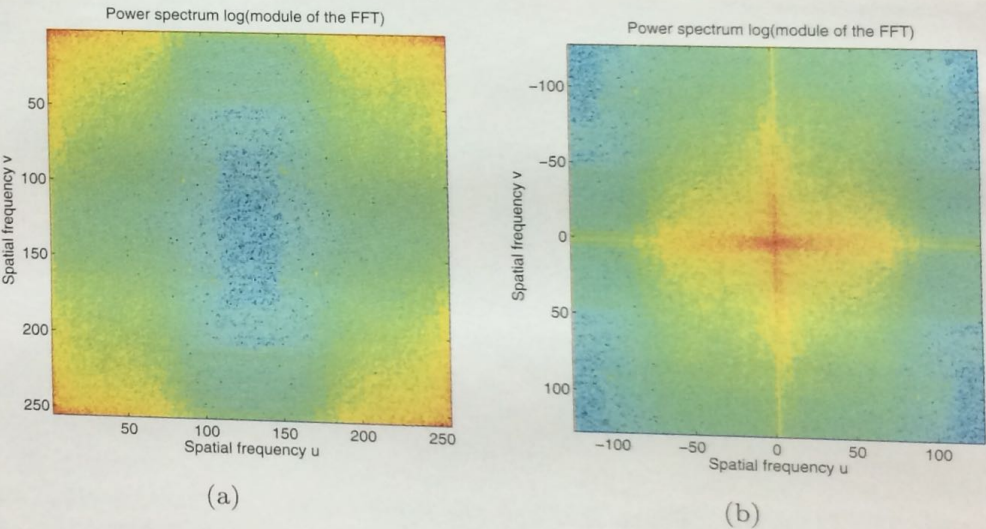


Plate 5: *Page 61, Figure 3.7.*

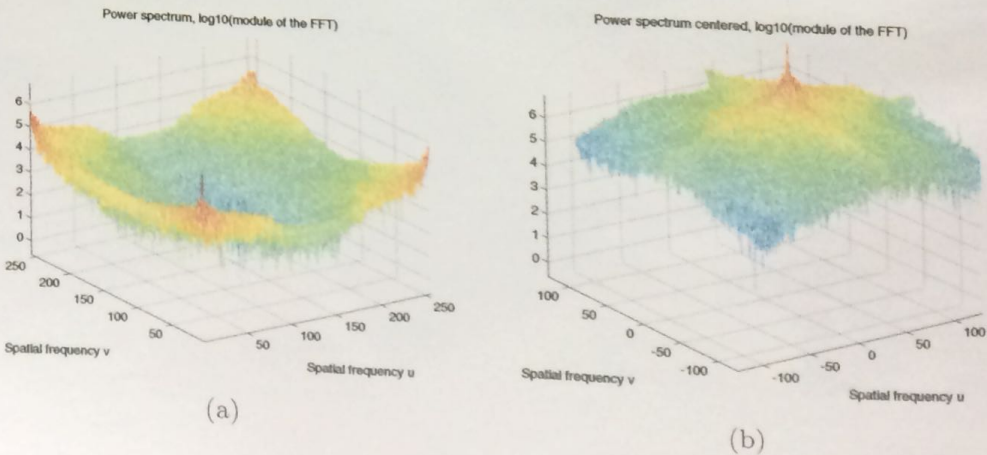


Plate 6: *Page 62, Figure 3.9.*